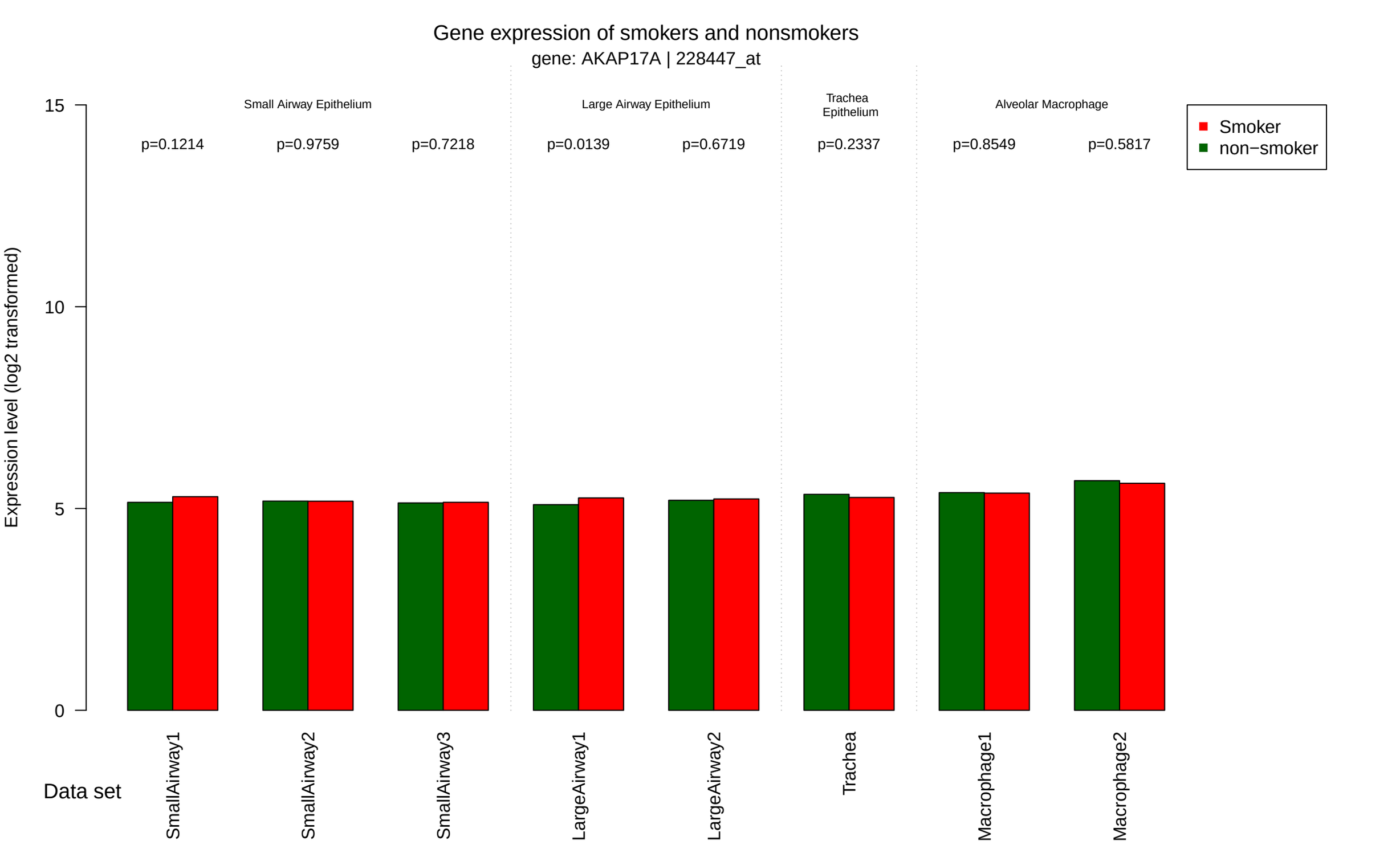
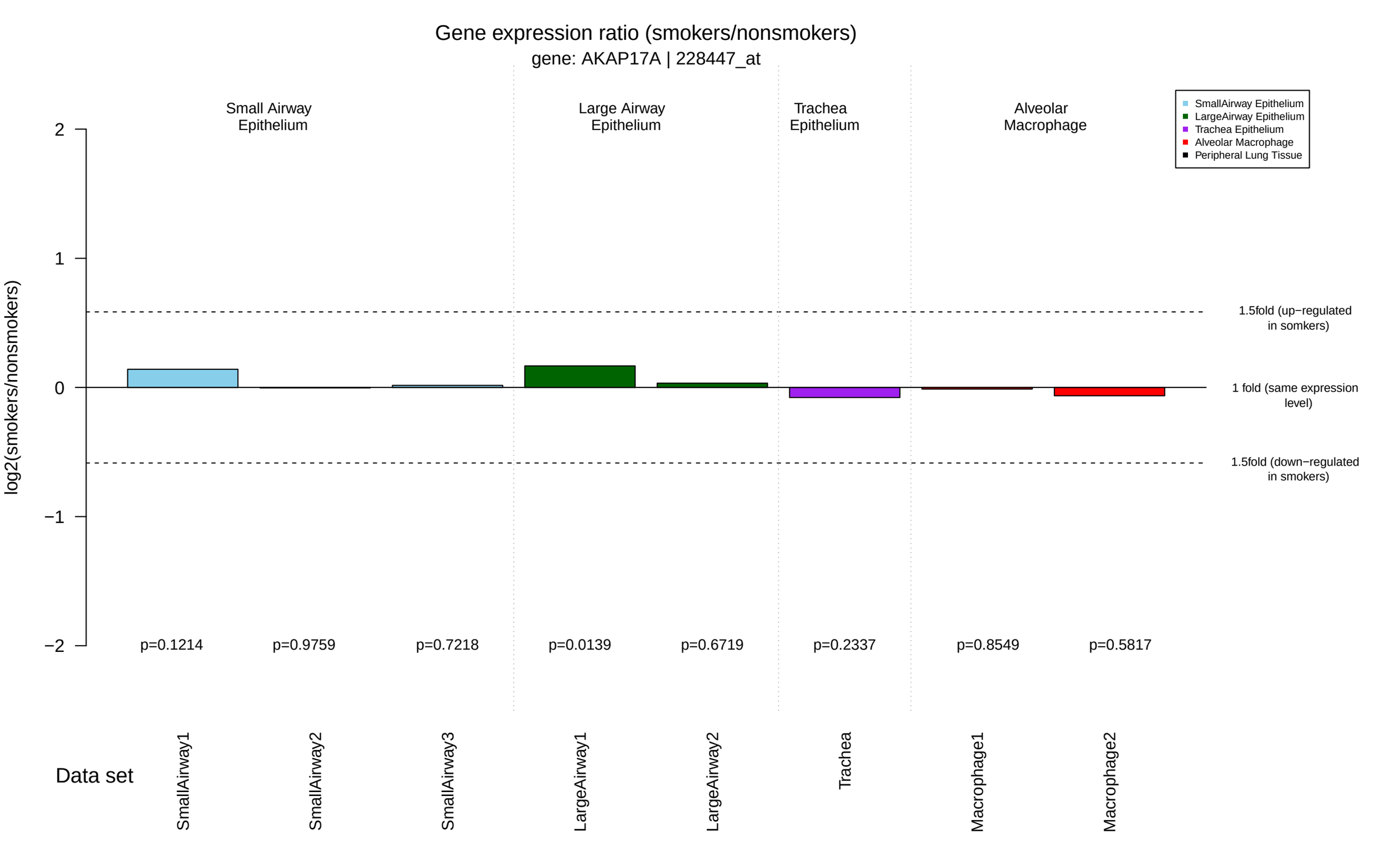
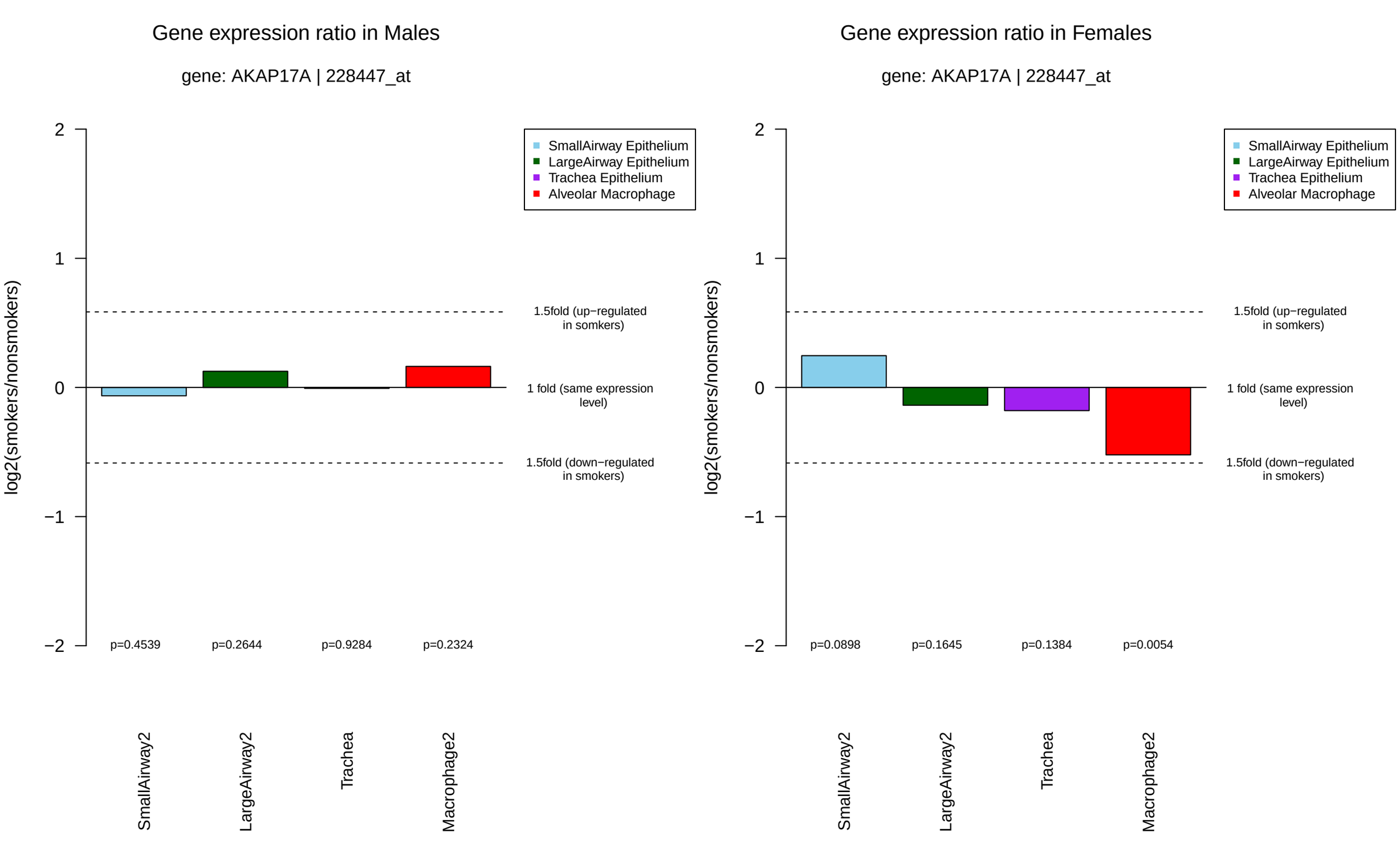
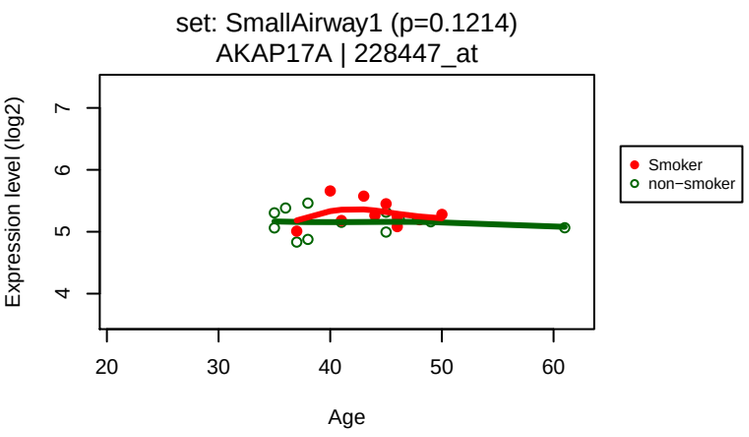
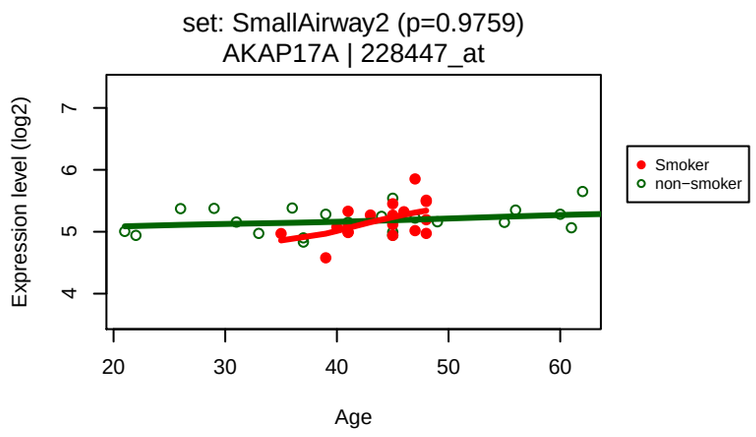
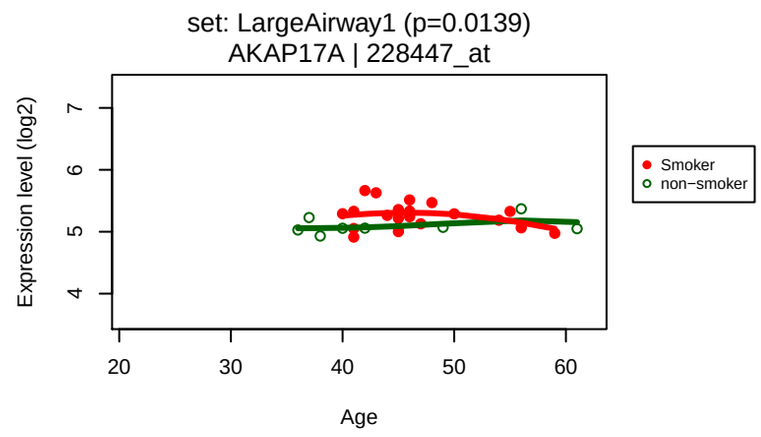

Gene expression of smokers and nonsmokers
gene: AKAP17A | 228447_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1214
p=0.9759
p=0.7218
p=0.0139
p=0.6719
p=0.2337
p=0.8549
p=0.5817
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

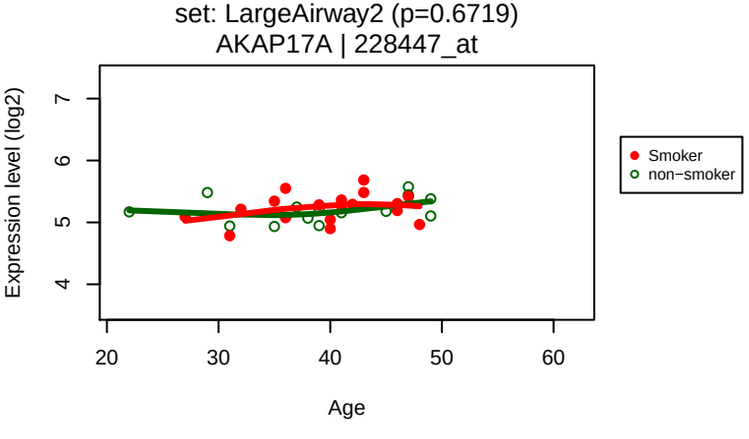

Gene expression ratio (smokers/nonsmokers)
gene: AKAP17A | 228447_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1214
p=0.9759
p=0.7218
p=0.0139
p=0.6719
p=0.2337
p=0.8549
p=0.5817
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

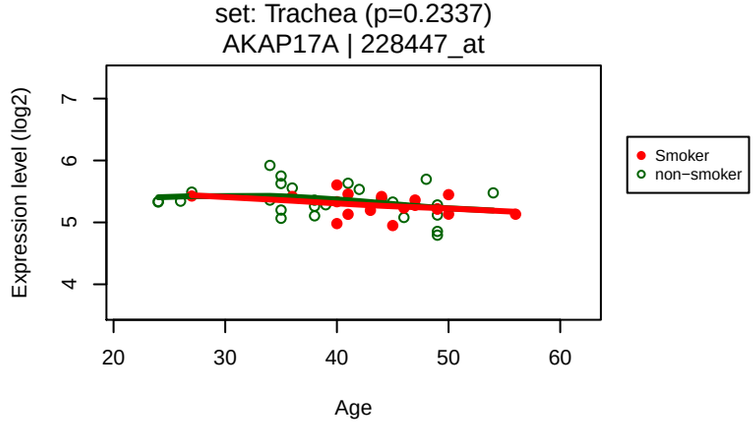

Gene expression ratio in Males
gene: AKAP17A | 228447_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4539
p=0.2644
p=0.9284
p=0.2324
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: AKAP17A | 228447_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0898
p=0.1645
p=0.1384
p=0.0054
SmallAirway2
LargeAirway2
Trachea
Macrophage2

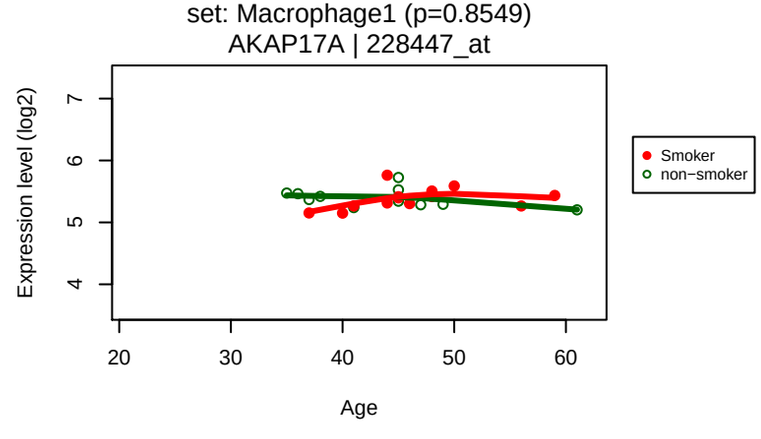

set: SmallAirway1 (p=0.1214)
AKAP17A | 228447_at
Expression level (log2)
Age
Smoker
non−smoker

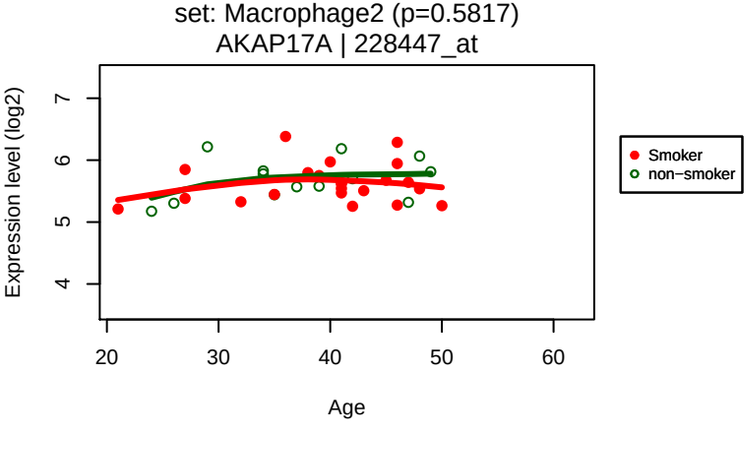

set: SmallAirway2 (p=0.9759)
AKAP17A | 228447_at
Expression level (log2)
Age
Smoker
non−smoker

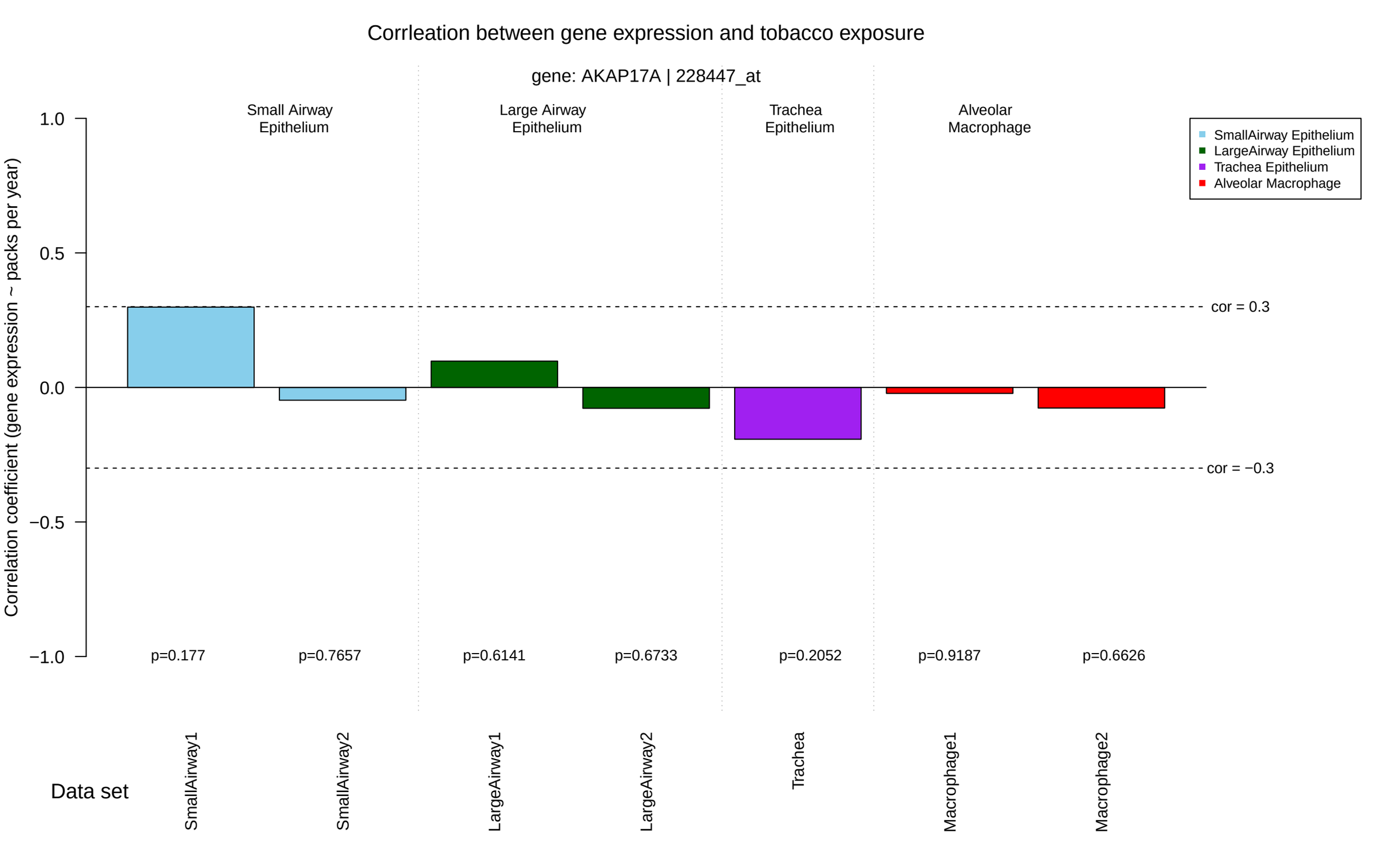

set: LargeAirway1 (p=0.0139)
AKAP17A | 228447_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6719)
AKAP17A | 228447_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2337)
AKAP17A | 228447_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8549)
AKAP17A | 228447_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5817)
AKAP17A | 228447_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: AKAP17A | 228447_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.177
p=0.7657
p=0.6141
p=0.6733
p=0.2052
p=0.9187
p=0.6626
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2